

ECE 3337: Test #1

Spring 2026

Last name of student POSTED SOLUTIONS	Student ID number _____
First name of student _____	Email address _____

Do This First:

1. Make sure that you have all the pages of the exam.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smart watch, ...)

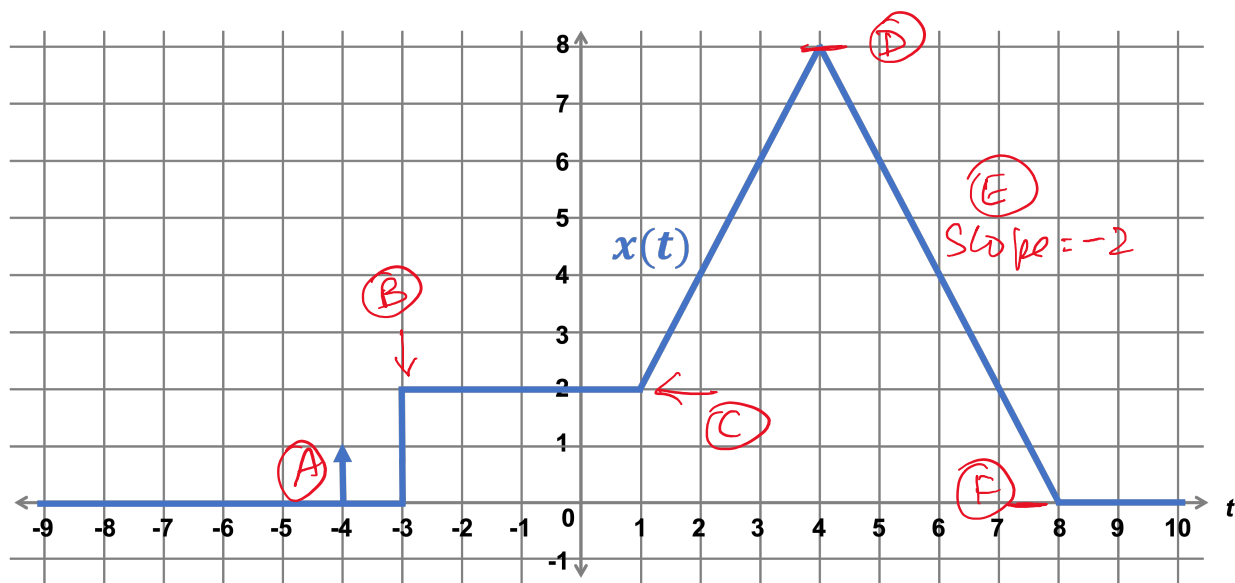
Exam Rules:

1. You are allowed to bring one 8.5" x 11" handwritten crib sheet (double-sided).
2. Indicate your answers in the spaces provided. Use the back of the page to carry out calculations.

Q1	Q2	Q3	Q4	Q5	Q6	Total
15	15	20	20	15	15	100
Waveforms & Transformations	Signal Properties	System Properties	Convolution	System Response	Frequency Response	_____
_____	_____	_____	_____	_____	_____	_____

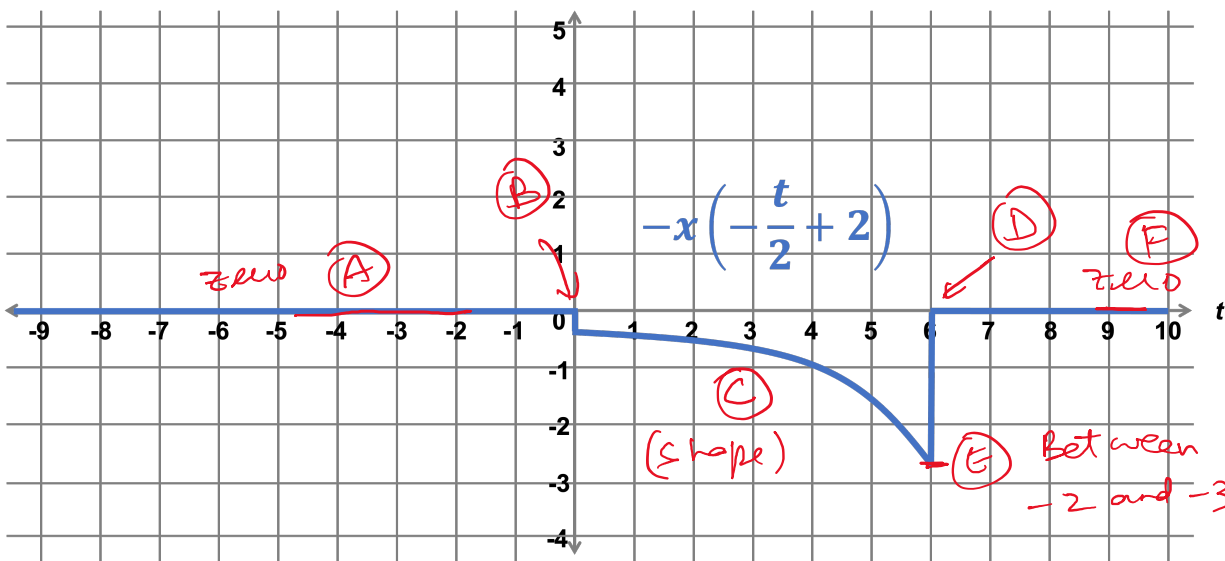
1. (15 points) Waveforms & Signal Transformations

a. (7 points) Plot the signal $x(t) = 2u(t + 3) + \delta(t + 4) + 2r(t - 1) - 4r(t - 4) + 2r(t - 8)$. Your final answer should be sketch below and will be the only response that will be graded.

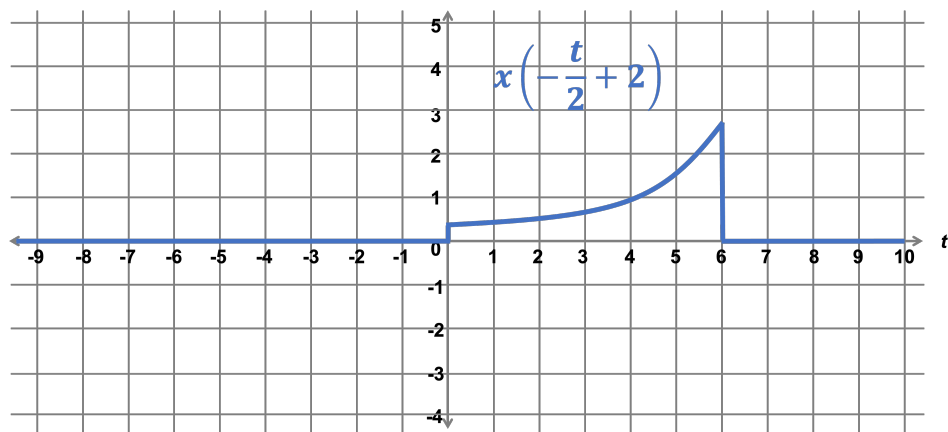
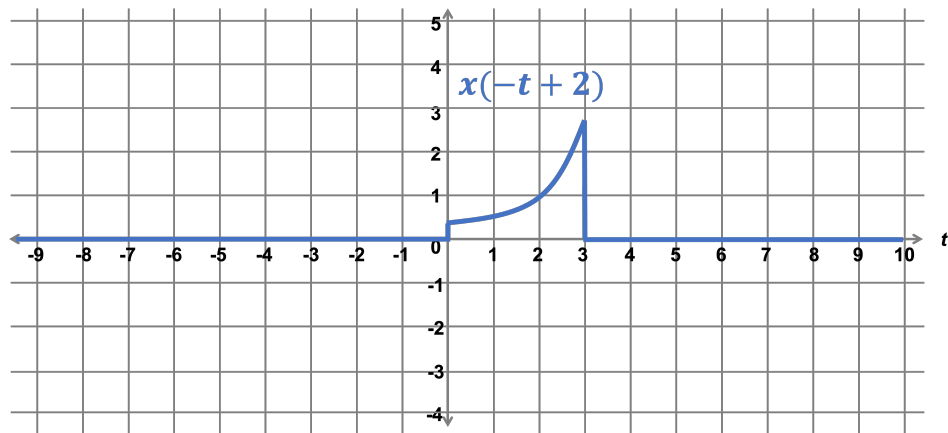
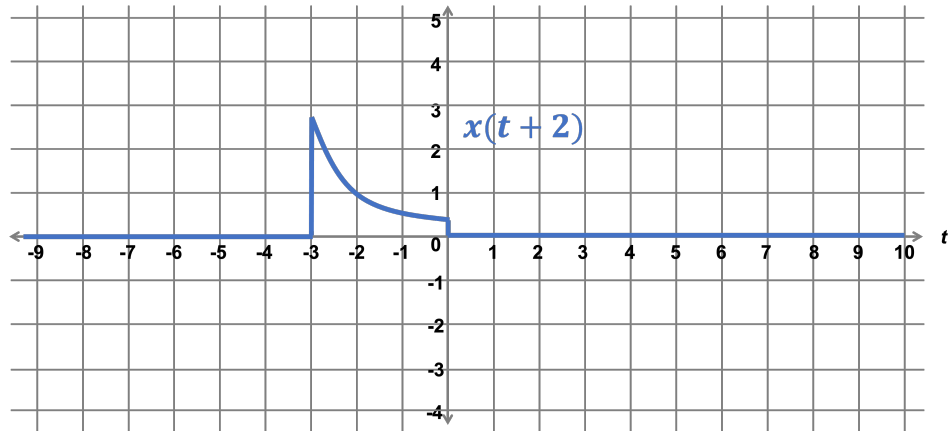
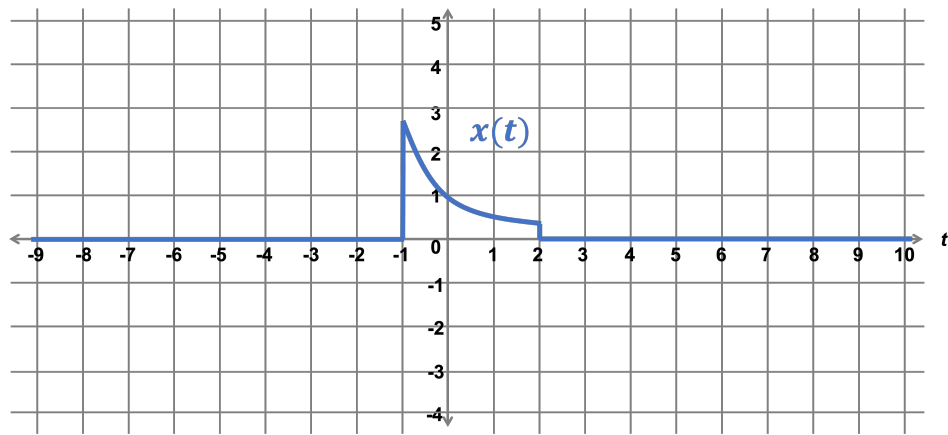


1 point for trying. 1 point for A through F.

b. (8 points) If $x(t) = e^{-t}[u(t + 1) - u(t - 2)]$, plot $y(t) = -x\left(-\frac{t}{2} + 2\right)$ on the graph below. Your final answer should be sketch below and will be the only response that will be graded.



2 points for trying. 1 point for A through F.



2. (15 points) Signal Properties

a. (4 points) A signal $x(t)$ has energy $E[x(t)]$. What is the energy of the signal $cx(at - b)$?

$$E[x(t)] = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E[cx(at-b)] = \int_{-\infty}^{\infty} |cx(at-b)|^2 dt = |c|^2 \int_{-\infty}^{\infty} |x(u)|^2 \frac{du}{a} = \frac{|c|^2}{a} \int_{-\infty}^{\infty} |x(u)|^2 du = \frac{|c|^2}{a} E[x(t)]$$

Let $u = at - b$
 $\Rightarrow du = a dt$

Grading: • 1 pt for trying

• +1 pt for (A)

• 4 pts correct answer

b. (4 points) Calculate the integral $I = \int_{-\infty}^{+\infty} t^5 \delta(3t - 5) dt$.

$$I = \int_{-\infty}^{\infty} t^5 \delta\left(3\left(t - \frac{5}{3}\right)\right) dt = \int_{-\infty}^{\infty} t^5 \underbrace{\frac{1}{3} \delta\left(t - \frac{5}{3}\right)}_{(A)} dt$$

(A): Time scaling of $\delta(\cdot)$

$$= \frac{1}{3} \int_{-\infty}^{\infty} t^5 \delta\left(t - \frac{5}{3}\right) dt = \frac{1}{3} \underbrace{\left(\frac{5}{3}\right)^5}_{(B)} \int_{-\infty}^{\infty} \delta\left(t - \frac{5}{3}\right) dt = \frac{1}{3} \cdot \left(\frac{5}{3}\right)^5 //$$

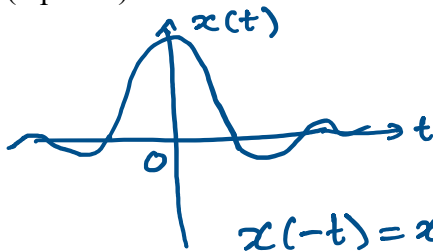
Grading: • 1 pt for trying

• +1 pt for (A) & (B) each

• 4 pts correct answer

(B): sampling property.

c. (3 points) Determine whether the signal $x(t) = \frac{\sin(3t)}{3t}$ has even, odd, or no symmetry.



$x(-t) = x(t) \Rightarrow$ Even symmetry

Grading:

• 1 pt for trying

• 3 pts correct answer

d. (4 points) Determine the amplitude, phase, and time period of the signal $x(t) = (3 + j4)e^{j3t+4}$.

$$x(t) = \sqrt{3^2+4^2} e^{j \tan^{-1}\left(\frac{4}{3}\right)} \times e^{j3t} \times e^4$$

Amplitude $= \sqrt{3^2+4^2} \times e^4 = 5e^4$ (A)

phase $= \tan^{-1}\left(\frac{4}{3}\right)$ (B)

Frequency $\omega = 3 = \frac{2\pi}{T_0}$

$\Rightarrow$ Period $T_0 = \frac{2\pi}{3}$ (C)

Grading: 1 pt for trying + 1 pt each correct (A), (B), (C).

3. (20 points) System Properties

a. (5 points) Does the system $y(t) = e^{-j 2x(t)} \sin(t)$ have the causality property? Explain your reasoning.

Notice that the output of the system $y(t)$ at time t depends only on the current value of the input $x(t)$. It does not depend on any future values of the input. Therefore, the system is causal.

(A)

(B)

1 point for trying. 2 points for A, B.

b. (5 points) Is the system $y(t) = \int_{t-1}^{t+3} x(\tau) d\tau$ time invariant? Explain your reasoning.

If the input is $x_1(t)$, the output is $y_1(t) = \int_{t-1}^{t+3} x_1(\tau) d\tau$.

(A)

Delaying this output by T seconds gives $y_1(t-T) = \int_{t-T-1}^{t-T+3} x_1(\tau) d\tau$.

If the input is $x_2(t) = x_1(t-T)$, then the output is

(B)

$$y_2(t) = \int_{t-1}^{t+3} x_1(\tau - T) d\tau$$

Substituting: $\tau - T = u$, the limits go from $t-1-T$ to $t+3-T$.

(C)

$$y_2(t) = \int_{t-T-1}^{t-T+3} x_1(u) du \stackrel{(C)}{=} y_1(t)$$

Therefore, the system is time invariant.

(D)

1 point for trying. 1 point for A-D.

c. (5 points) Is the system $y(t) = 10 e^{j|x(t)|^2}$ BIBO stable? Explain your reasoning.

The system is BIBO stable if the output $|y(t)| = |10 e^{j|x(t)|^2}|$ is bounded when the input $|x(t)|$ is bounded. If $|x(t)| < C$ is bounded, then $|x(t)|^2 < C^2$ is also bounded. Note that $|e^{j\theta}| = 1$ for all finite values of θ .

(A)

Consequently, $|y(t)| = |10 e^{j|x(t)|^2}| = 10 |e^{j|x(t)|^2}| = 10$ is bounded by 10 if $x(t)$ is bounded. Therefore, the system is BIBO stable.

(B)

(C)

1 point for trying. 1 point for A-B. 2 points for C.

d. (5 points) Is the system defined by $y(t) = \frac{\sin(100t)}{2t} x(t)$ linear? Explain your answer.

Let the outputs for inputs $x_1(t)$ and $x_2(t)$ be

$$y_1(t) = \frac{\sin(100t)}{2t} x_1(t) \text{ and}$$

$$y_2(t) = \frac{\sin(100t)}{2t} x_2(t)$$

The output for $x(t) = a x_1(t) + b x_2(t)$ is

(A)

$$y(t) = \frac{\sin(100t)}{2t} x(t)$$

$$y(t) = \frac{\sin(100t)}{2t} [a x_1(t) + b x_2(t)]$$

(B)

$$= a \frac{\sin(100t)}{2t} x_1(t) + b \frac{\sin(100t)}{2t} x_2(t)$$

$$= a y_1(t) + b y_2(t)$$

Therefore, the system is linear.

(D)

(C)

1 point for trying. 1 point for A-D.

4. (20 points) Convolution Integral

a. (5 points) Compute the following convolution without computing any integrals

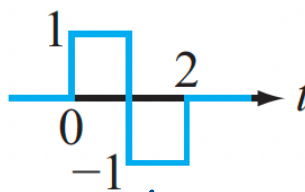
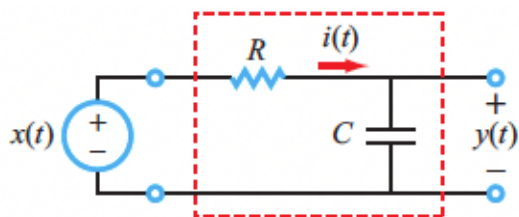
$y(t) = u(t+1) * [\delta(t-2) + u(t-3) + r(t-1)]$, where $r(t)$ is the ramp function.

$$y(t) = u(t+1) * \delta(t-2) + u(t+1) * u(t-3) + u(t+1) * (t-1)u(t-1)$$

$$= \underbrace{u(t-1)}_{(A)} + \underbrace{r(t-2)}_{(B)} + \underbrace{\frac{1}{2}t^2 u(t)}_{(C)}$$

Grading: 2 pts for trying
+ 1 pt each correct (A), (B), (C).

b. (7 points) Write down the step response $y_{step}(t)$ of the following circuit with $RC = 1$, and zero initial conditions. Then calculate the response to the input waveform indicated below in terms of $y_{step}(t)$.



$$x(t) = u(t) - 2u(t-1) + u(t-2) \quad (A)$$

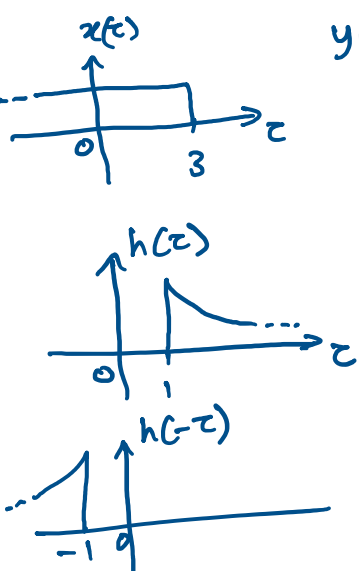
$$\Rightarrow y(t) = y_{step}(t) - 2y_{step}(t-1) + y_{step}(t-2) \quad (B) \quad (C) \quad (D)$$

where

$$y_{step}(t) = (1 - e^{-t})u(t) \quad (E)$$

Grading: 2 pts for trying
+ 1 pt each correct (A), (B), (C), (D), (E).

c. (8 points) Using explicit integration and showing all steps, calculate the output $y(t)$ of an LTI system with impulse response $h(t) = e^{-2t}u(t-1)$ for the input $x(t) = u(3-t)$.



$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \quad \text{(A)}$$

choose

$$= \int_{-\infty}^{\infty} u(3-\tau) e^{-2(t-\tau)} u(t-\tau-1) d\tau \quad \text{(B)}$$

$3-\tau > 0$
 $\Rightarrow \tau < 3$

$t-\tau-1 > 0$
 $\Rightarrow \tau < t-1$

} (C)

when $t < 4 \Rightarrow$ limits are $-\infty$ to $t-1$ I_1
when $t > 4 \Rightarrow$ limits are $-\infty$ to 3 I_2

$$I_1 = u(4-t) \int_{-\infty}^{t-1} e^{-2(t-\tau)} d\tau, \quad I_2 = u(t-4) \int_{-\infty}^3 e^{-2(t-\tau)} d\tau$$

$$= u(4-t) e^{-2t} \int_{-\infty}^{t-1} e^{2\tau} d\tau + u(t-4) e^{-2t} \int_{-\infty}^3 e^{2\tau} d\tau$$

(D) (E)

$$= u(4-t) e^{-2t} \left\{ \frac{e^{2(t-1)}}{2} - 0 \right\} + u(t-4) e^{-2t} \left\{ \frac{e^6}{2} - 0 \right\}$$

$$= \frac{u(4-t)}{2} e^{-2} + \frac{u(t-4)}{2} e^{-2t+6} //$$

(F)

Grading:

- 2 pts for trying
- +1 pt each correct (A) - (F)
- Full points for correct final answer.

5. (15 points) System Response

Consider the LTI system given by the differential equation

$$3 \frac{dy(t)}{dt} + 4y(t) = -\frac{dx(t)}{dt}.$$

Assume that $y(0^-) = 0$.

- a. (7 points) Find the step response $y_{step}(t)$ of the above system.

To find the step response, give the system an input $x(t) = u(t)$. — (A)

$$\frac{dx(t)}{dt} = \delta(t) \quad \text{— (B)}$$

The differential equation reduces to

$$3 \frac{dy(t)}{dt} + 4y(t) = -\delta(t).$$

$$\frac{dy(t)}{dt} + \frac{4}{3}y(t) = -\frac{1}{3}\delta(t). \quad \text{— (C)}$$

This is a first-order linear differential equation. The integrating factor is $e^{\int \frac{4}{3} dt} = e^{\frac{4}{3}t}$. Multiplying by the integrating factor and simplifying,

$$\frac{dy(t)}{dt} e^{\frac{4}{3}t} + \frac{4}{3}y(t)e^{\frac{4}{3}t} = -\frac{1}{3}\delta(t)e^{\frac{4}{3}t} = -\frac{1}{3}\delta(t)$$

$$\frac{d}{dt} \left(y(t)e^{\frac{4}{3}t} \right) = -\frac{1}{3}\delta(t) \quad \text{— (D)}$$

Integrating,

$$y(t)e^{\frac{4}{3}t} \Big|_{0^-}^t = -\int_{0^-}^t \frac{1}{3}\delta(t) dt = -\frac{1}{3} \text{ for } t > 0 \text{ and } 0 \text{ otherwise.}$$

$$y(t)e^{\frac{4}{3}t} - y(0^-) = -\frac{1}{3}u(t)$$

$$y(t)e^{\frac{4}{3}t} = -\frac{1}{3}u(t)$$

$$y(t) = -\frac{1}{3}e^{-\frac{4}{3}t} \text{ for } t > 0 \text{ and } 0 \text{ otherwise.}$$

Therefore, $y(t) = -\frac{1}{3}e^{-\frac{4}{3}t}u(t)$ — (E)

2 points for trying. 1 point for A-E.

- b. (3 points) Find the impulse response $h(t)$ of the above system.

To find the impulse response, we can differentiate the step response.

$$h(t) = \frac{d}{dt} y_{step}(t) \quad \text{--- (A)}$$

$$= \frac{4}{9} e^{-\frac{4}{3}t} u(t) - \frac{1}{3} \delta(t) \quad \text{--- (B)}$$

1 point for trying. 1 point for A-B.

- c. (5 points) Calculate the frequency response $\hat{H}(\omega)$ for the above system.

Since the system is LTI, let $x(t) = e^{j\omega t}$ and $y(t) = \hat{H}(\omega)e^{j\omega t}$. (A)

Therefore, $\frac{d}{dt} x(t) = j\omega e^{j\omega t}$, $\frac{d}{dt} y(t) = j\omega \hat{H}(\omega)e^{j\omega t}$. (B)

$3 \frac{dy(t)}{dt} + 4y(t) = -\frac{dx(t)}{dt}$ reduces to

$$3j\omega \hat{H}(\omega)e^{j\omega t} + 4 \hat{H}(\omega)e^{j\omega t} = -j\omega e^{j\omega t} \rightarrow \text{(C)}$$

$$\hat{H}(\omega)(3j\omega + 4) = -j\omega$$

$$\hat{H}(\omega) = \frac{-j\omega}{(3j\omega + 4)} \rightarrow \text{(D)}$$

1 point for trying. 1 point for A-D.

6. (15 points) System Frequency Response

- a. (5 points) Determine the value(s) of constant B that make the following system BIBO stable and critically damped:

$$\frac{d^2y}{dt^2} + B \frac{dy}{dt} + 9y(t) = 5 \frac{dx}{dt} + 4x(t)$$

1. For BIBO stability, $B > 0$ (A)
 2. For critical damping, $B^2 - 4 \cdot 1 \cdot 9 = 0 \Rightarrow B = \pm 6$ (B)
- To meet both criteria, $B = +6$ //

Grading: • 1 pt for trying
• +1 pt (A), (B)
• 5 pts for correct final answer

- b. (5 points) A linear time-invariant system is described by the following differential equation:

$$\frac{d^2y}{dt^2} - 8 \frac{dy}{dt} + 15y(t) = \frac{dx}{dt} + x(t)$$

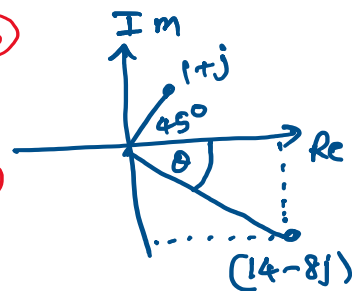
Derive the frequency response function $\hat{H}(\omega)$ of this system, and calculate the output for the input $x(t) = \cos(t)$.

$$\hat{H}(\omega) = \frac{(j\omega) + 1}{(j\omega)^2 - 8(j\omega) + 15} \quad (A)$$

when $x(t) = \cos(t)$ for which $\omega = 1$, $\hat{H}(\omega) = \frac{j+1}{j^2 - 8j + 15} = \frac{j+1}{-8j+14}$

$$|\hat{H}(\omega)| = \left| \frac{j+1}{-8j+14} \right| = \frac{\sqrt{2}}{\sqrt{8^2+14^2}} = \frac{\sqrt{2}}{\sqrt{196+64}} = \frac{\sqrt{2}}{\sqrt{260}} = \frac{1}{\sqrt{130}} \quad (B)$$

$$\angle \hat{H}(\omega) = \angle(j+1) - \angle(-8j+14) = 45^\circ - \tan^{-1}\left(\frac{-8}{14}\right) \quad (C)$$



$$\Rightarrow y(t) = \frac{1}{\sqrt{130}} \cos\left(t + 45^\circ + \tan^{-1}\left(\frac{8}{14}\right)\right) \quad (D)$$

Grading: • 1 pt for trying
• +1 pt each (A) - (D)
• Full points for correct final answer

c. (5 points) The following input-output pair is observed for a linear time-invariant system:

$$x(t) = 1 + 3 \cos(t - 45^\circ) + 4 \sin(2t) \rightarrow 4 \cos(2t - 90^\circ)$$

$$y(t) = 7 + 2 \cos(t - 30^\circ) + \cos(2t - 45^\circ)$$

Calculate the output for the new input $x_2(t) = 5 + \sin(t) + \cos(2t)$.

$$\downarrow$$

$$\cos(t - 90^\circ)$$

For $\omega = 0$, $|\hat{H}(\omega)| = 7$, phase $= 0^\circ$ (A)

$\omega = 1$ $|\hat{H}(\omega)| = \frac{2}{3}$, phase $= 15^\circ$ (B)

$\omega = 2$ $|\hat{H}(\omega)| = \frac{1}{4}$, phase $= 45^\circ$ (C)

$$\Rightarrow y_2(t) = 5 \times 7 + \frac{2}{3} \cos(t - 90^\circ + 15^\circ) + \frac{1}{4} \cos(2t + 45^\circ), \text{ (D)}$$

Grading: 1 pt for trying
+ 1 pt (A) - (D)
Full points correct final answer